

Computer Science & IT

Discrete Mathematics



Mathematical Logic

Lecture No. 06



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Recap of Previous Lecture



Topic

Negation of predicate formula



Topic

Sentences with multiple quantifiers



Topic

Relationship diagram



Topics to be Covered



✓ **Topic**

Logical equivalences and Logical implications

✓ **Topic**

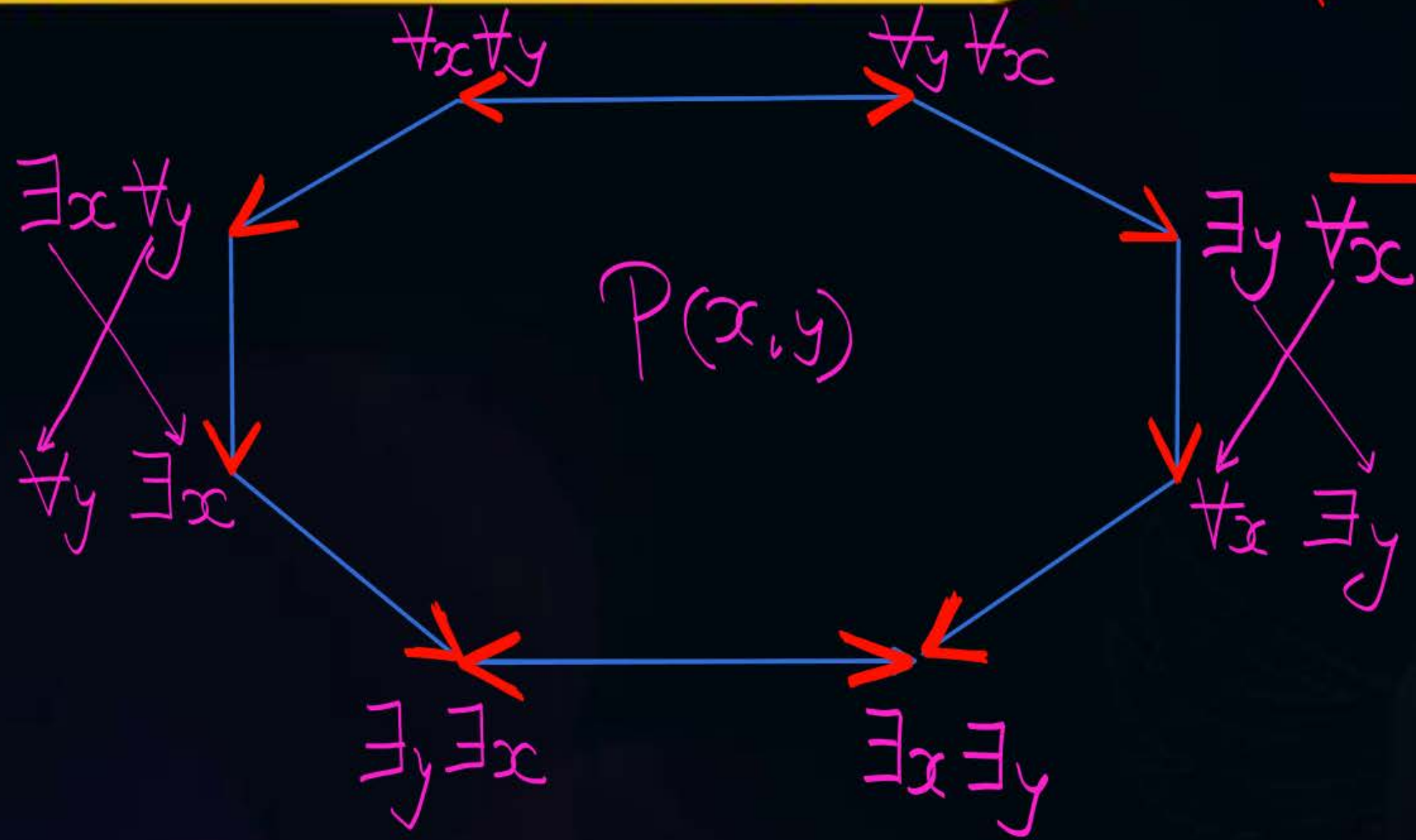
Rules of inference

✓ **Topic**

Tautology w.r.t. Predicate logic



Topic : Relationship diagram



$\longleftrightarrow$ it will represent Equivalence

$\longrightarrow$ it will represent logical implication

Today's Topic



Topic : Rules of Inferences w.r.t. predicate logic

- ❑ In predicate logic, we can use some additional inference rules, along with all the rules of inference we have discussed in propositional logic.
- ❑ Additional Rules of Inference w.r.t. Predicate Logic
 1. Universal Instantiation (Universal Specification)
 2. Universal Generalization
 3. Existential Instantiation (Existential Specification)
 4. Existential Generalization



Topic : Universal Instantiation

Universal specification



$$\forall x \{ P(x) \}$$

is valid
◦◦ $P(a)$ is true for any 'a' in the domain
for all ^{or} 'a' in the domain



Topic : Universal Generalization

$P(a)$ is true for all 'a' in domain

$$\therefore \forall x \{P(x)\}$$

We can use any variable
for generalization

{ i.e. $\forall z \{P(z)\}$ is also
Valid }



Topic : Existential Instantiation

$$\exists x \{ P(x) \}$$

$\exists x [P(x)]$ is valid

o.o $P(a)$ is true for some 'a' in domain



Topic : Existential Generalization

$P(a)$ is true for some (at least one) 'a' in domain

$$\therefore \exists z \{P(z)\}$$

$\forall x \{P(x)\}$ will be invalid

Equivalences & Implications

H.W

$$\begin{array}{ccc} \textcircled{1} & \forall x \{ P(x) \wedge Q(x) \} & \begin{array}{c} \xrightarrow{\textcircled{1}} \\ \equiv \\ \xleftarrow{\textcircled{2}} \end{array} & \{ \forall x \{ P(x) \} \wedge \forall x \{ Q(x) \} \} \end{array}$$

Diagram illustrating the equivalence between two logical expressions:

- Expression 1: $\forall x \{ P(x) \wedge Q(x) \}$
- Expression 2: $\{ \forall x \{ P(x) \} \wedge \forall x \{ Q(x) \} \}$

The expressions are shown to be equivalent ($\equiv$) with arrows indicating the direction of the proof steps (1 and 2). Each arrow is accompanied by a checkmark.

Equivalences & Implications

H.W.

$$\textcircled{2} \quad \exists x \{ P(x) \vee Q(x) \} \begin{array}{c} \xrightarrow{\textcircled{1}} \\ \equiv \\ \xleftarrow{\textcircled{2}} \end{array} \{ \exists x \{ P(x) \} \vee \exists x \{ Q(x) \} \}$$

Equivalences & Implications

HW

$$\textcircled{3} \quad \forall x \{ P(x) \vee Q(x) \} \quad \begin{array}{c} \textcircled{1} \xrightarrow{\text{red X}} \\ \equiv \text{red X} \\ \xleftarrow{\text{green}} \end{array} \quad \textcircled{2} \quad \{ \forall x \{ P(x) \} \vee \forall x \{ Q(x) \} \}$$

Equivalences & Implications

HW

$$\textcircled{4} \quad \exists x \{ P(x) \wedge Q(x) \} \overset{\textcircled{1}}{\implies} \checkmark \quad \{ \exists x \{ P(x) \} \wedge \exists x \{ Q(x) \} \}$$

~~$\equiv$~~

$$\overset{\textcircled{2}}{\impliedby} \quad \{ \exists x \{ P(x) \} \wedge \exists x \{ Q(x) \} \}$$



Topic : Equivalences & Implications



⑥

$$\forall x \{P(x) \leftrightarrow Q(x)\} \equiv (\forall x \{P(x)\} \leftrightarrow \forall x \{Q(x)\})$$

1. $\forall x [P(x) \wedge Q(x)] \equiv [\forall x P(x)] \wedge [\forall x Q(x)]$

2. $\exists x [P(x) \vee Q(x)] \equiv [\exists x P(x)] \vee [\exists x Q(x)]$

3. $[\forall x P(x)] \vee [\forall x Q(x)] \Rightarrow \forall x [P(x) \vee Q(x)]$

4. $\exists x [P(x) \wedge Q(x)] \Rightarrow [\exists x P(x)] \wedge [\exists x Q(x)]$

5. $\forall x [P(x) \rightarrow Q(x)] \Rightarrow [\forall x P(x)] \rightarrow [\forall x Q(x)]$



Topic : Some important equivalences

1. $\forall x [P(x) \wedge Q] \equiv \forall x P(x) \wedge Q$

2. $\exists x [P(x) \vee Q] \equiv \exists x P(x) \vee Q$

3. $\forall x [P(x) \vee Q] \equiv \forall x P(x) \vee Q$

4. $\exists x [P(x) \wedge Q] \equiv \exists x P(x) \wedge Q$

Note: In the given formulae
'Q' is a predicate formula
without any free occurrence
of 'x'

$\Downarrow$
i.e. Truth value of Q
does not vary with 'x'

$\Downarrow$
i.e. if Q is true, then it
will remain true for all x
if Q is false then it will
remain false for all x



Topic : Some important equivalences

1. $\forall x [P(x) \wedge Q] \equiv \forall x P(x) \wedge Q$

Case ① $Q = \text{False}$

When Q is false, the both LHS and R.H.S. will always return false for all values of x .

i.e. When $Q = \text{False}$, then same truth table for LHS & RHS

2. $\exists x [P(x) \vee Q] \equiv \exists x P(x) \vee Q$

Case ② $Q = \text{True}$

① For LHS to be true, 'P' must be true for all x , and in that case $\forall x \{P(x)\}$ will also return true. Hence RHS will also be true.

② For RHS to be true $\forall x \{P(x)\}$ must be true, Hence for all 'x' $P(x) \wedge Q$ will also be true, Hence LHS will also be true.

3. $\forall x [P(x) \vee Q] \equiv \forall x P(x) \vee Q$

4. $\exists x [P(x) \wedge Q] \equiv \exists x P(x) \wedge Q$



Topic : Some important equivalences

1. $\forall x [P(x) \wedge Q] \equiv \forall x P(x) \wedge Q$

2. $\exists x [P(x) \vee Q] \equiv \exists x P(x) \vee Q$

3. $\forall x [P(x) \vee Q] \equiv \forall x P(x) \vee Q$

4. $\exists x [P(x) \wedge Q] \equiv \exists x P(x) \wedge Q$



Topic : Some important equivalences

5. $\forall x [P \rightarrow Q(x)] \equiv P \rightarrow \forall x Q(x)$

6. $\exists x [P \rightarrow Q(x)] \equiv P \rightarrow \exists x Q(x)$

7. $\forall x [P(x) \rightarrow Q] \equiv \exists x P(x) \rightarrow Q$

8. $\exists x [P(x) \rightarrow Q] \equiv \forall x P(x) \rightarrow Q$

In first two predicate formula, 'P' is a predicate formula without any free occurrence of 'x'.

In 7th and 8th predicate formula 'Q' is the predicate free formula without any occurrence of 'x'.



Topic : Some important equivalences

5. $\forall x [P \rightarrow Q(x)] \equiv P \rightarrow \forall x Q(x)$

No Change

6. $\exists x [P \rightarrow Q(x)] \equiv P \rightarrow \exists x Q(x)$

No Change

7. $\forall x [P(x) \rightarrow Q] \equiv \exists x P(x) \rightarrow Q$

Change

8. $\exists x [P(x) \rightarrow Q] \equiv \forall x P(x) \rightarrow Q$

Change



Topic : Some important equivalences

5. $\forall x [P \rightarrow Q(x)] \equiv P \rightarrow \forall x Q(x)$

$$\forall x \{ \underbrace{\sim P} \vee \underbrace{Q(x)} \} \equiv \underbrace{\sim P} \vee \underbrace{\forall x \{ Q(x) \}} \equiv P \rightarrow \forall x \{ Q(x) \} = \text{RHS}$$

6. $\exists x [P \rightarrow Q(x)] \equiv P \rightarrow \exists x Q(x)$

$$\exists x \{ \underbrace{\sim P} \vee \underbrace{Q(x)} \} \equiv \underbrace{\sim P} \vee \underbrace{\exists x \{ Q(x) \}} \equiv P \rightarrow \exists x \{ Q(x) \} = \text{RHS}$$

7. $\forall x [P(x) \rightarrow Q] \equiv \exists x P(x) \rightarrow Q$

$$\begin{aligned} \forall x \{ \underbrace{\sim P(x)} \vee \underbrace{Q} \} &\equiv \forall x \{ \underbrace{\sim P(x)} \} \vee \underbrace{Q} \\ &\equiv \underbrace{\sim(\exists x \{ P(x) \})} \vee \underbrace{Q} \equiv \exists x \{ P(x) \} \rightarrow Q = \text{RHS} \end{aligned}$$

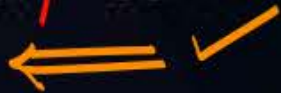
8. $\exists x [P(x) \rightarrow Q] \equiv \forall x P(x) \rightarrow Q$

$$\begin{aligned} \exists x \{ \underbrace{\sim P(x)} \vee \underbrace{Q} \} &\equiv \underbrace{\exists x \{ \sim P(x) \}} \vee \underbrace{Q} \equiv \underbrace{\sim \forall x \{ P(x) \}} \vee \underbrace{Q} \equiv \forall x \{ P(x) \} \rightarrow Q = \text{RHS} \\ &\equiv \sim \forall x \{ P(x) \} \end{aligned}$$

#Q. Let $P(x)$ and $Q(x)$ be arbitrary predicates. Which of the following statements is always TRUE?

~~A~~

$$(\forall x(P(x) \vee Q(x))) \Rightarrow ((\forall x P(x)) \vee (\forall x Q(x)))$$



B

$$(\forall x(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x P(x)) \Rightarrow (\forall x Q(x)))$$

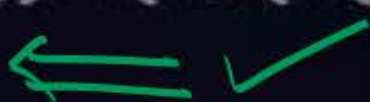


~~C~~

$$(\forall x (P(x) \Rightarrow \forall x Q(x))) \Rightarrow (\forall x (P(x) \Rightarrow Q(x)))$$

~~D~~

$$((\forall x(P(x) \Leftrightarrow (\forall x Q(x)))) \Rightarrow (\forall x(P(x) \Leftrightarrow Q(x))))$$



#Q. Which of the following predicate calculus statements is/are valid

☒ **A**

$$(\forall x)P(x) \vee (\forall x)Q(x) \rightarrow (\forall x)\{P(x) \vee Q(x)\}$$

☒ **B**

$$(\exists x)P(x) \wedge (\exists x)Q(x) \rightarrow (\exists x)\{P(x) \wedge Q(x)\}$$

☒ **C**

$$(\exists x)\{P(x) \vee Q(x)\} \rightarrow (\forall x)P(x) \vee (\forall x)Q(x)$$

{valid only if domain is non-empty}

☒ **D**

$$\underline{(\exists x)\{P(x) \vee Q(x)\}} \rightarrow \{\exists x P(x) \vee (\forall x)Q(x)\}$$

#Q. Which of the following predicate calculus statements is/are valid

☒ **A**

$$\forall(x)(P(x) \Rightarrow Q(x)) \Rightarrow ((\forall(x)P(x)) \Rightarrow (\forall(x)Q(x)))$$

☐ **B**

$$\exists x(P(x) \vee Q(x)) \Rightarrow ((\exists xP(x)) \Rightarrow (\exists xQ(x)))$$

$\begin{matrix} \text{True} & \longrightarrow & \text{False} \end{matrix}$

$= \text{False}$

☐ **C**

$$\exists x(P(x) \wedge Q(x)) \Leftrightarrow ((\exists xP(x)) \wedge (\exists xQ(x)))$$

$\begin{matrix} \text{True} & \longrightarrow & \text{True} \\ \text{False} & \longrightarrow & \text{False} \end{matrix}$

☐ **D**

$$\forall(x)\exists y P(x, y) \Leftrightarrow \exists y \forall(x)P(x, y)$$

$\begin{matrix} \text{True} & \longrightarrow & \text{False} \\ \text{False} & \longrightarrow & \text{True} \end{matrix}$

#Q. Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

i.e. $\alpha = P(x)$

$\beta = Q(x)$

~~A~~

$$((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]) \Rightarrow (\forall x)[\alpha \Rightarrow \beta]$$

$$\{ \forall x \{ P(x) \Rightarrow \forall x \{ Q(x) \} \} \Rightarrow \forall x \{ P(x) \rightarrow Q(x) \}$$

~~B~~

$$((\forall x)[\alpha] \Rightarrow (\exists x)[\alpha \wedge \beta])$$

$$\forall x \{ P(x) \} \Rightarrow \exists x \{ P(x) \wedge Q(x) \}$$

} if $\alpha = \text{True}$ for all x

$\wedge \beta = \text{False}$ for all x , then LHS = True but RHS = False

~~C~~

$$((\forall x)[\alpha \vee \beta] \Rightarrow (\exists x)[\alpha]) \Rightarrow (\forall x)[\alpha]$$

$$\emptyset \rightarrow \text{True} \rightarrow \text{False}$$

Let $\alpha = \text{False}$ for some x but not for all

~~D~~

$$((\forall x)[\alpha \Rightarrow \beta] \Rightarrow ((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]))$$

$$\forall x \{ P(x) \rightarrow Q(x) \} \Rightarrow \{ \forall x \{ P(x) \} \rightarrow \forall x \{ Q(x) \} \}$$

#Q. Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first-order formula with x as a free variable and β is a first-order formula with no free variable.

A $[\beta \rightarrow (\exists x, \alpha(x))] \rightarrow [\forall x, \beta \rightarrow \alpha(x)]$

B $[\exists x, \beta \rightarrow (\alpha(x))] \rightarrow [\beta \rightarrow (\forall x, \alpha(x))]$

C $[(\exists x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$

D $[(\forall x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$

H.W.

Check whether other three implications are valid or not?

[MSQ]



#Q. Which one of the following predicate formulae is NOT logically valid? Note that W is a predicate formula without any free occurrence of x .

A

$$\exists x(p(x) \wedge \underline{W}) \equiv (\exists x p(x)) \wedge W \quad \{ \text{valid} \}$$

B

$$\forall x(p(x) \rightarrow \underline{W}) \equiv (\forall x p(x)) \rightarrow \underline{W} \quad \{ \text{invalid} \}$$

it has to $\exists x$

C

$$\exists x(p(x) \rightarrow \underline{W}) \equiv (\forall x p(x)) \rightarrow \underline{W} \quad \{ \text{valid} \}$$

D

$$\underline{\forall x(p(x) \vee W)} \equiv (\forall x p(x)) \vee \underline{W} \quad \{ \text{valid} \}$$

[MSQ]

it is a general statement, i.e. we will use all.



#Q. Which one the following is the most appropriate logical formula to represent the statement "gold and silver ornaments are precious". The following notations are used:

$G(x)$: x is a gold ornament

$S(x)$: x is a silver ornament

$P(x)$: x is precious

$\wedge$ or $\vee$

A

$$\forall x (P(x) \rightarrow G(x) \wedge S(x))$$

All precious things are made of gold & silver

B

$$\exists x ((G(x) \wedge S(x)) \rightarrow P(x))$$

At least one ornament is made of mixture of gold & silver and it is precious

C

$$\forall x ((G(x) \wedge S(x)) \rightarrow P(x))$$

All ornaments that are made of mixture of gold & silver are precious

D

$$\forall x ((G(x) \vee S(x)) \rightarrow P(x))$$

it will be wrong, because an ornament is either of gold or silver but not both.

HW
#Q. What is the correct translation of the following statement into mathematical logic?

“some real numbers are rational”

A $\exists x (real(x) \vee rational(x))$

B $\exists x (real(x) \wedge rational(x))$

C $\forall x (real(x) \rightarrow rational(x))$

D $\forall x (rational(x) \rightarrow real(x))$

#Q. What is the logical translation of the following statement?

"None of my friends are perfect"

- A** $\exists x (F(x) \wedge \sim P(x))$
- B** $\exists x (\sim F(x) \wedge \sim P(x))$
- C** $\exists x (\sim F(x) \wedge P(x))$
- D** $\sim (\exists x (F(x) \wedge P(x)))$

H.W.
[MSQ]

#Q. The CORRECT formula of the sentence "not all rainy days are cold"

- A** $\forall d (rainy(d) \wedge \sim cold(d))$
- B** $\exists d (\sim rainy(d) \rightarrow cold(d))$
- C** $\forall d (\sim rainy(d) \rightarrow cold(d))$
- D** $\exists d (rainy(d) \wedge \sim cold(d))$

[MSQ]

#Q. Which one of the first order predicate calculus statements given below correctly expresses the following English statements? "Tigers and lions attack if they are hungry or threatened".

- A** $\forall x[(tiger(x) \wedge lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$
- B** $\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \wedge attacks(x))]$
- C** $\forall x[(tiger(x) \wedge lion(x)) \rightarrow (attacks(x) \rightarrow (hungry(x) \vee threatened(x)))]$
- D** $\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$

#Q. Which of the following argument is valid?

are

A

$\forall x (P(x))$ follows from the premises $\exists x (P(x) \wedge Q(x))$

B

$\exists x (P(x) \wedge Q(x))$ follows from the premises $\exists x (P(x)) \wedge \exists x (Q(x))$

C

$\exists x (Q(x))$ follows from the premises $\{\forall x (P(x) \Rightarrow Q(x)), \exists y (Q(y))\}$

D

$\sim P(a, b)$ follows from the premises $\{\forall x \forall y \{ P(x, y) \rightarrow W(x, y) \}, \sim W(a, b)\}$

~~H.W.~~
#Q. Check whether the following argument is true or false?

$\forall x\{F(x) \rightarrow \sim S(x)\}$ follows from the premises

$\{ \exists x(F(x) \wedge S(x)) \rightarrow \forall y(M(y) \rightarrow W(y)) , \exists y(M(y) \wedge \sim W(y)) \}$

Q: Which of the following predicate formula is/are valid

✓ (a) $\exists x \{P(x) \wedge Q(x)\} \implies (\exists x \{P(x)\} \wedge \exists x \{Q(x)\})$

✓ (b) $\exists x \{P(x) \rightarrow Q(x)\} \implies \exists x \{\sim P(x) \vee Q(x)\}$

✓ (c) $\sim \forall x \{P(x) \wedge Q(x)\} \iff \exists x \{P(x) \rightarrow \sim Q(x)\}$

✓ (d) $\exists x \forall y (P(x, y)) \implies \forall y \exists x (P(x, y))$

Q: Which of the following predicate formula is/are tautology

(a) $\exists x \{P(x) \wedge Q(x)\} \implies (\exists x \{P(x)\} \wedge \exists x \{Q(x)\})$

(b) $\exists x \{P(x) \rightarrow Q(x)\} \implies \exists x \{\sim P(x) \vee Q(x)\}$

(c) $\sim \forall x \{P(x) \wedge Q(x)\} \iff \exists x \{P(x) \rightarrow \sim Q(x)\}$

(d) $\exists x \forall y (P(x, y)) \implies \forall y \exists x (P(x, y))$

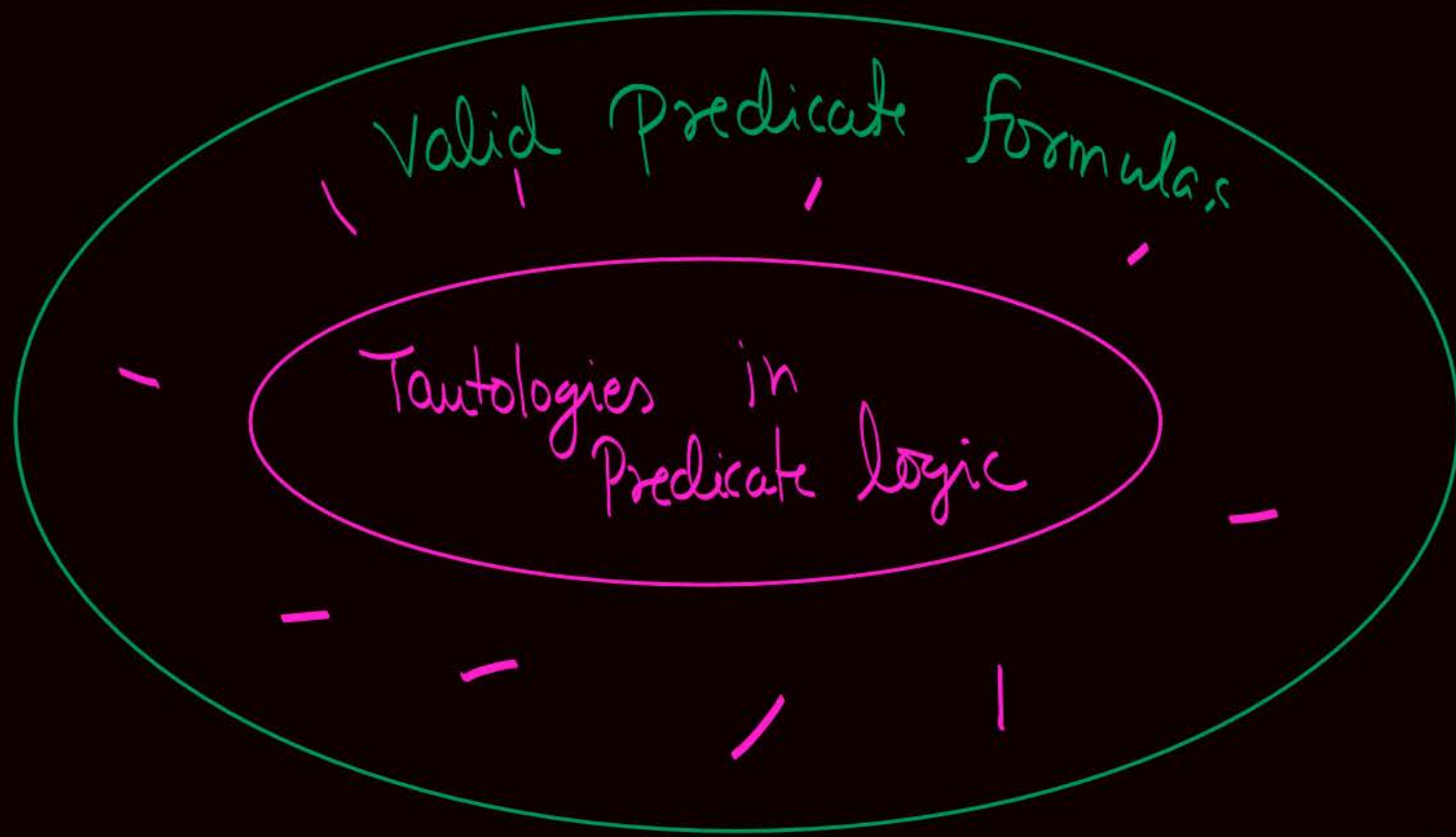


Topic : Tautology in predicate logic

- ✦ In propositional logic, tautologies and validities are same.
- ✦ A propositional function which is always true is called valid propositional function or tautology.

But in predicate logic, distinction is maintained between logical validities and tautologies.

A predicate formula which is always true is called a valid predicate formula but it may or may not be a tautology. In predicate logic tautologies are proper subset of logical validities.



→ Every tautology in predicate logic is also a valid Predicate formula but Every valid Predicate formula need not be a tautology.

→ If a predicate formula is invalid, then it can never be a tautology.



Topic : Tautology in predicate logic

- A tautology in predicate logic is a sentence that can be obtained by taking a tautology of propositional logic and uniformly replacing each propositional variable by a predicate formula (one formula per propositional variable).

NOTE: To check whether a given predicate formula is a tautology or not we can only use the concepts that we have learned in propositional logic,

i.e., we can not use any concept of predicate logic

eg.
$$P \longrightarrow (Q \longrightarrow P)$$

let $P = \forall x \{P(x)\}$

& $Q = \forall x \{Q(x)\}$

} then

$$\forall x \{P(x)\} \longrightarrow (\forall x \{Q(x)\} \longrightarrow \forall x \{P(x)\})$$

is a tautology {it will be valid as well}

Q: Which of the following predicate formula is/are tautology

- X (a) $\exists x \{P(x) \wedge Q(x)\} \Rightarrow (\exists x \{P(x)\} \wedge \exists x \{Q(x)\}) \equiv P \rightarrow Q \wedge R$
 Not a tautology
- ✓ (b) $\exists x \{P(x) \rightarrow Q(x)\} \Rightarrow \exists x \{ \sim P(x) \vee Q(x) \} \equiv P \rightarrow P$
 We know that $P(x) \rightarrow Q(x)$ is a tautology from propositional logic
- X (c) $\sim \forall x \{P(x) \wedge Q(x)\} \Leftrightarrow \exists x \{P(x) \rightarrow \sim Q(x)\} \equiv P \Leftrightarrow Q$
 $P \Leftrightarrow Q$ is not a tautology
- X (d) $\exists x \forall y (P(x,y)) \Rightarrow \forall y \exists x (P(x,y)) \equiv P \rightarrow Q$
 $P \rightarrow Q$ is not a tautology



2 mins Summary



✓
Topic

Logical equivalences and Logical implications

✓
Topic

Rules of inference

✓
Topic

Tautology w.r.t. Predicate logic

THANK - YOU